

Data & AI

9. Convolutional Neural Networks

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Last Time

Assignment 1: ResNet18, trained again on images you collect

- Batches of 32, one step per batch
- ResNet18 is a convolutional neural network

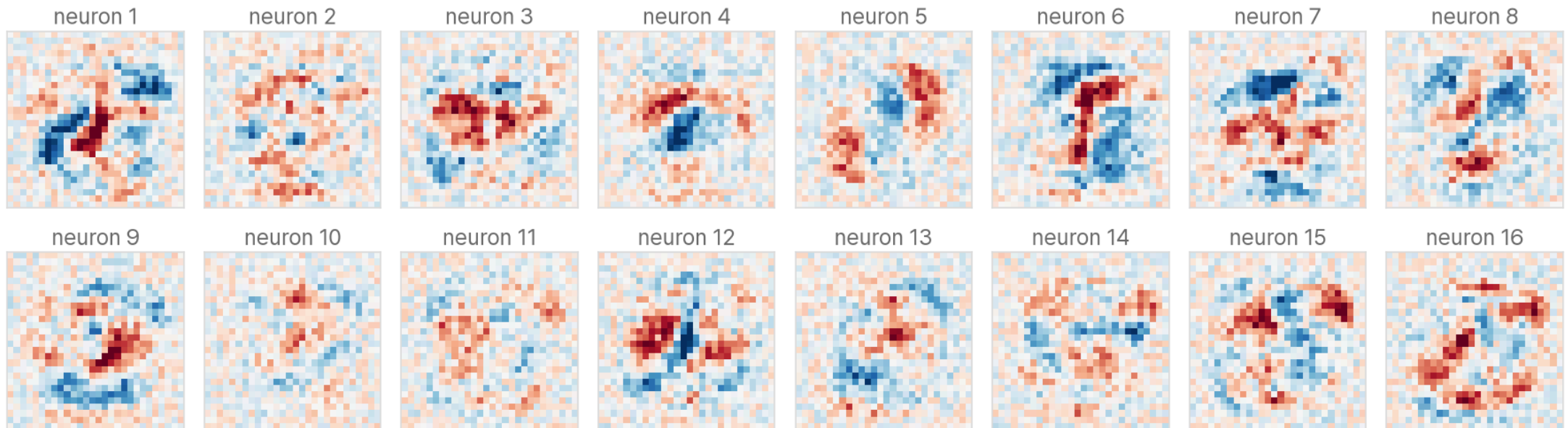
Today

- What goes wrong when an image is flattened
- Convolution: one small filter applied at every position
- Pooling, and stacking layers into a CNN
- ResNet18, the model in Assignment 1

Flattening an image

Hidden Neurons of the Digit Classifier

MLP from lecture 7: one weight per pixel, drawn as 28×28 (red positive, blue negative)

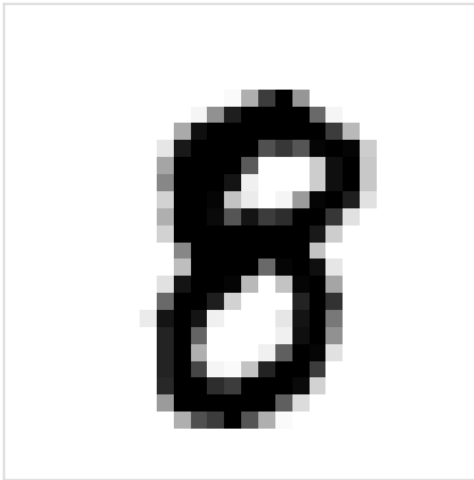


- Each neuron looks at the whole image at once
- Each weight belongs to one fixed pixel position

Shifting the Digit

The trained MLP, the same test image moved to the right

shifted 0 px right



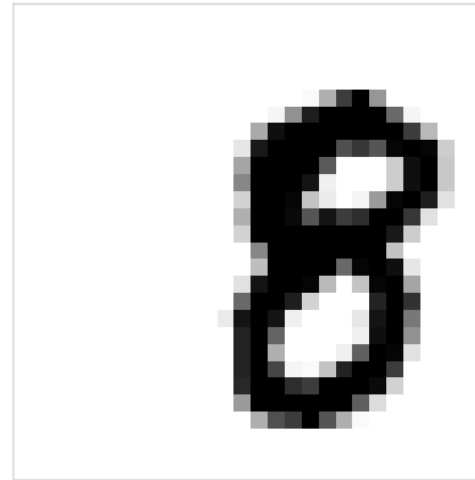
answer 8 $p(8) = 0.96$

shifted 2 px right



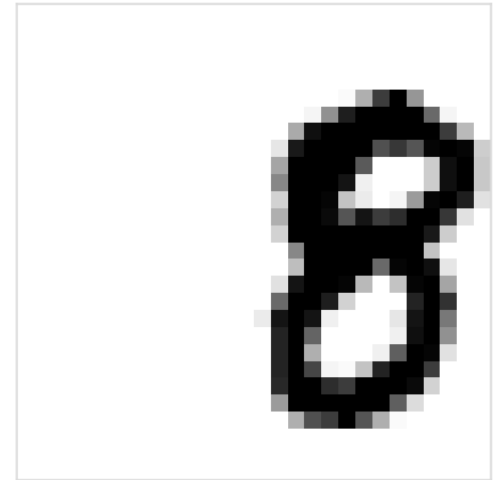
answer 8 $p(8) = 0.82$

shifted 4 px right



answer 3 $p(8) = 0.03$

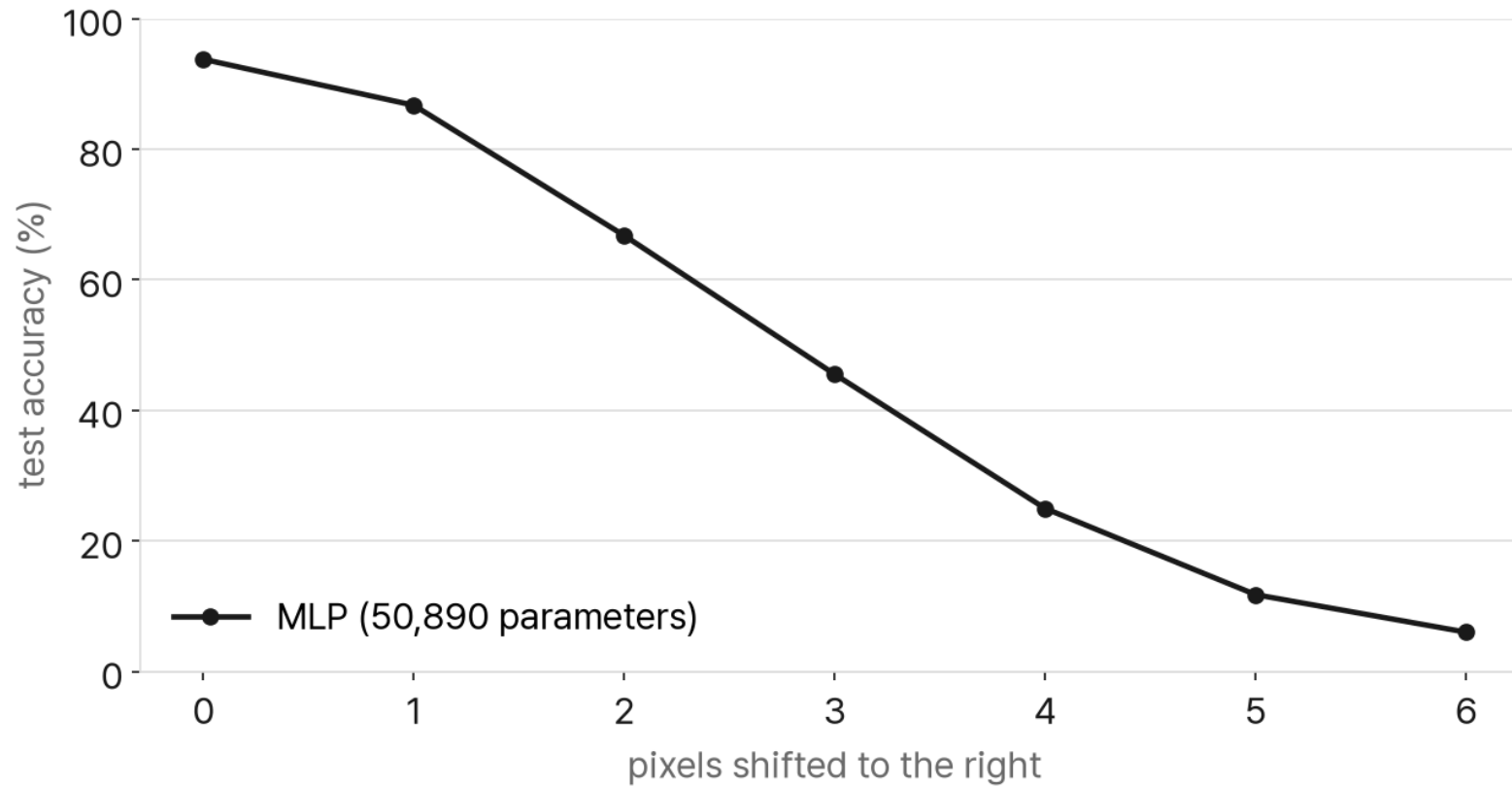
shifted 6 px right



answer 7 $p(8) = 0.00$

Test Accuracy After Shifting

All 2,000 test images, moved the same number of pixels



Photos Are Larger

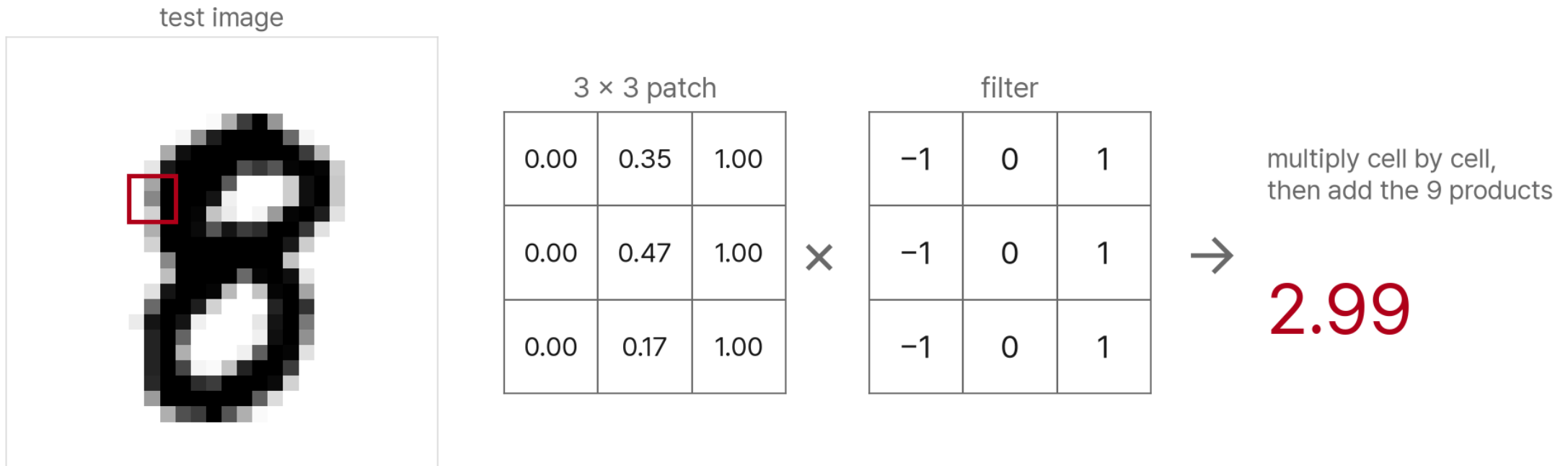
A 224×224 colour photo, as in Assignment 1

- $224 \times 224 \times 3 = 150,528$ input numbers
- One layer of 64 hidden neurons: $150,528 \times 64 = 9,633,792$ weights
- And every weight still belongs to one pixel position

Convolution

A Filter

A small grid of weights, here 3×3

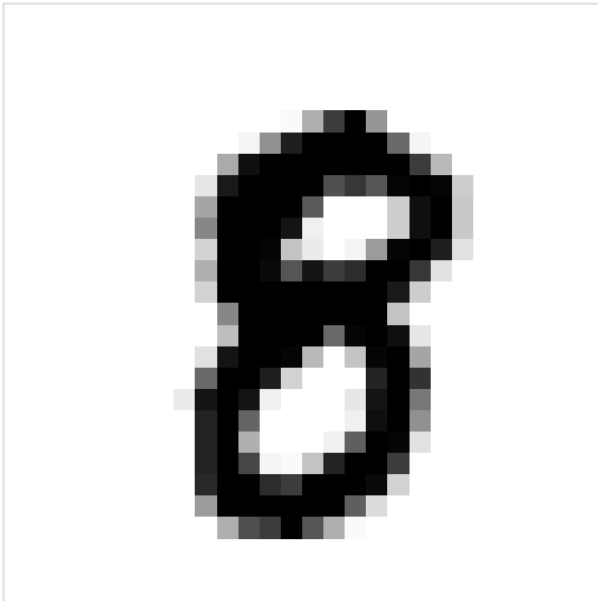


- Place it on one 3×3 patch of the image
- Multiply cell by cell and add: one number

Every Position

Slide the same filter over the whole image: a feature map

test image



filter (vertical edge)

-1	0	1
-1	0	1
-1	0	1



feature map, 28 × 28



Different Filters, Different Maps

vertical edge

-1	0	1
-1	0	1
-1	0	1

horizontal edge

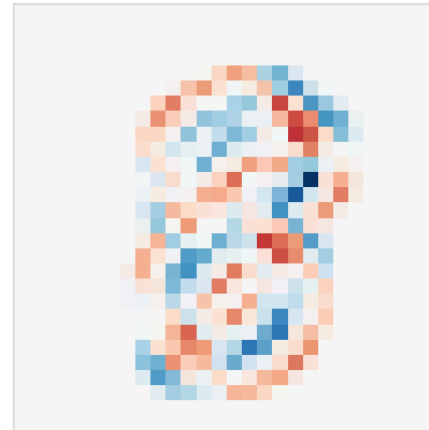
-1	-1	-1
0	0	0
1	1	1

diagonal \

2	-1	-1
-1	2	-1
-1	-1	2

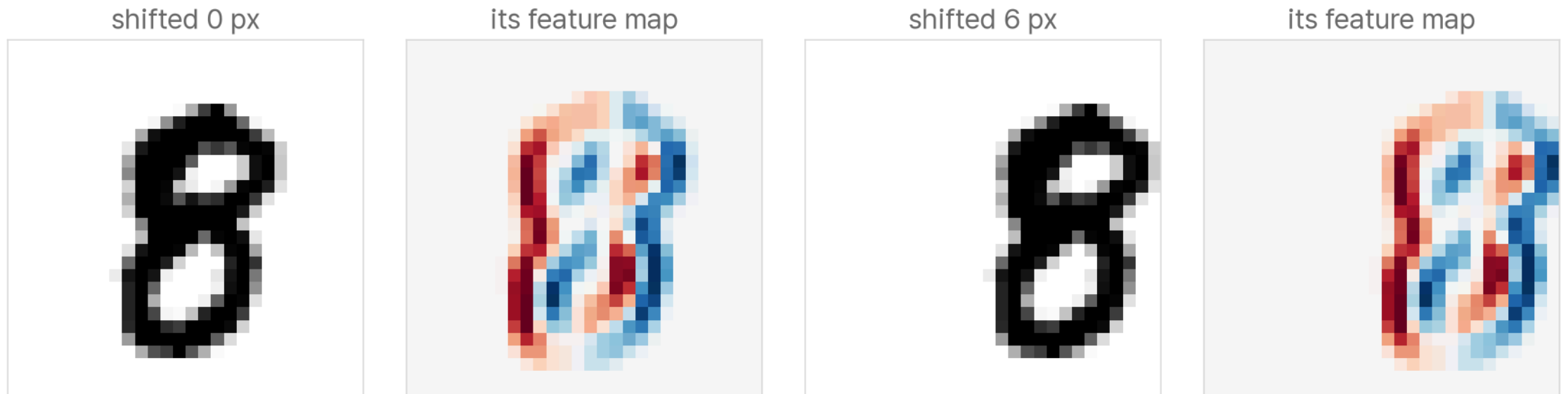
diagonal /

-1	-1	2
-1	2	-1
2	-1	-1



The Same Filter Everywhere

Move the digit, and the feature map moves with it



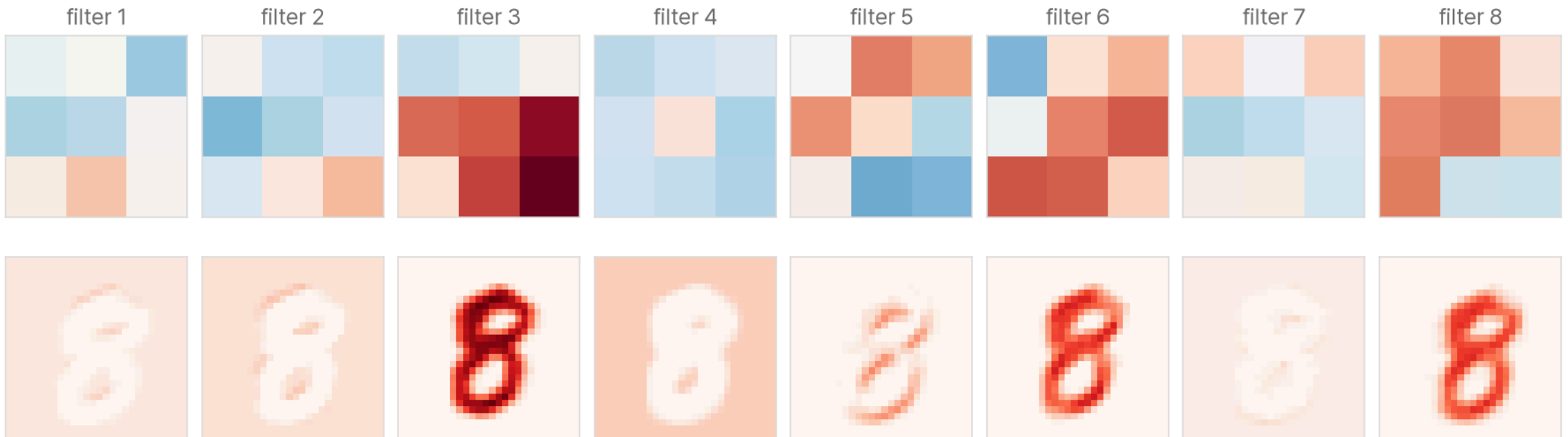
- 9 weights, used at all 784 positions
- A stroke is found wherever it is

Filters Are Learned

- The 9 numbers of each filter are parameters
- Found by gradient descent, as in lectures 4 and 6
- Nobody writes "vertical edge" into the network

Learned Filters

First layer of a small CNN trained on MNIST: 8 filters and their maps for one test image



Many Filters, Many Channels

- A layer has many filters: 8 filters give 8 feature maps
- Output shape: $8 \times 28 \times 28$, 8 channels
- A colour image already has 3 channels: R, G, B
- A filter covers all input channels: $3 \times 3 \times 3$ for colour

Pooling and stacking

Max Pooling

Keep the largest value in each 2×2 block: height and width halve

4×4 of the map (after ReLU)

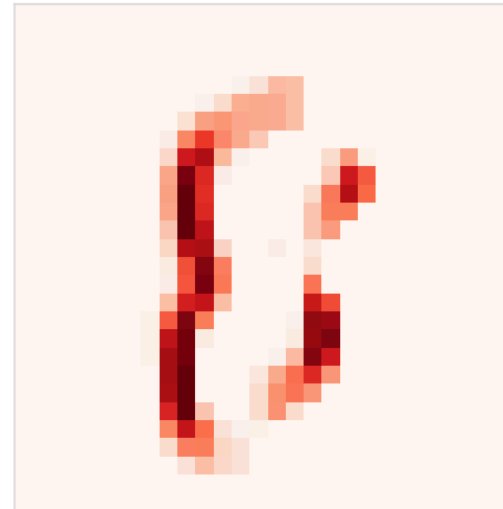
0.5	2.2	2.5	0.8
0.9	2.9	2.1	0.1
1.0	3.0	2.0	0.0
0.9	3.0	2.1	0.0



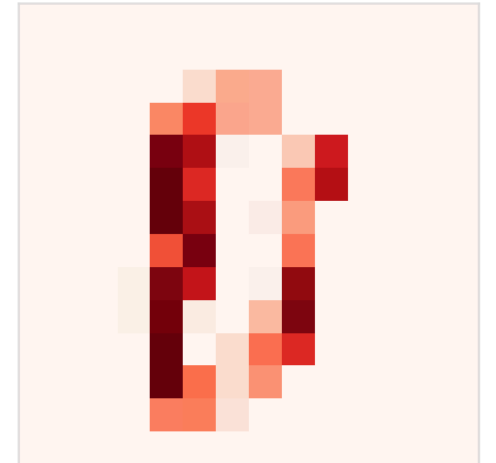
largest in each 2×2

2.9	2.5
3.0	2.1

28×28



14×14



A Small CNN for MNIST

Shapes in the network

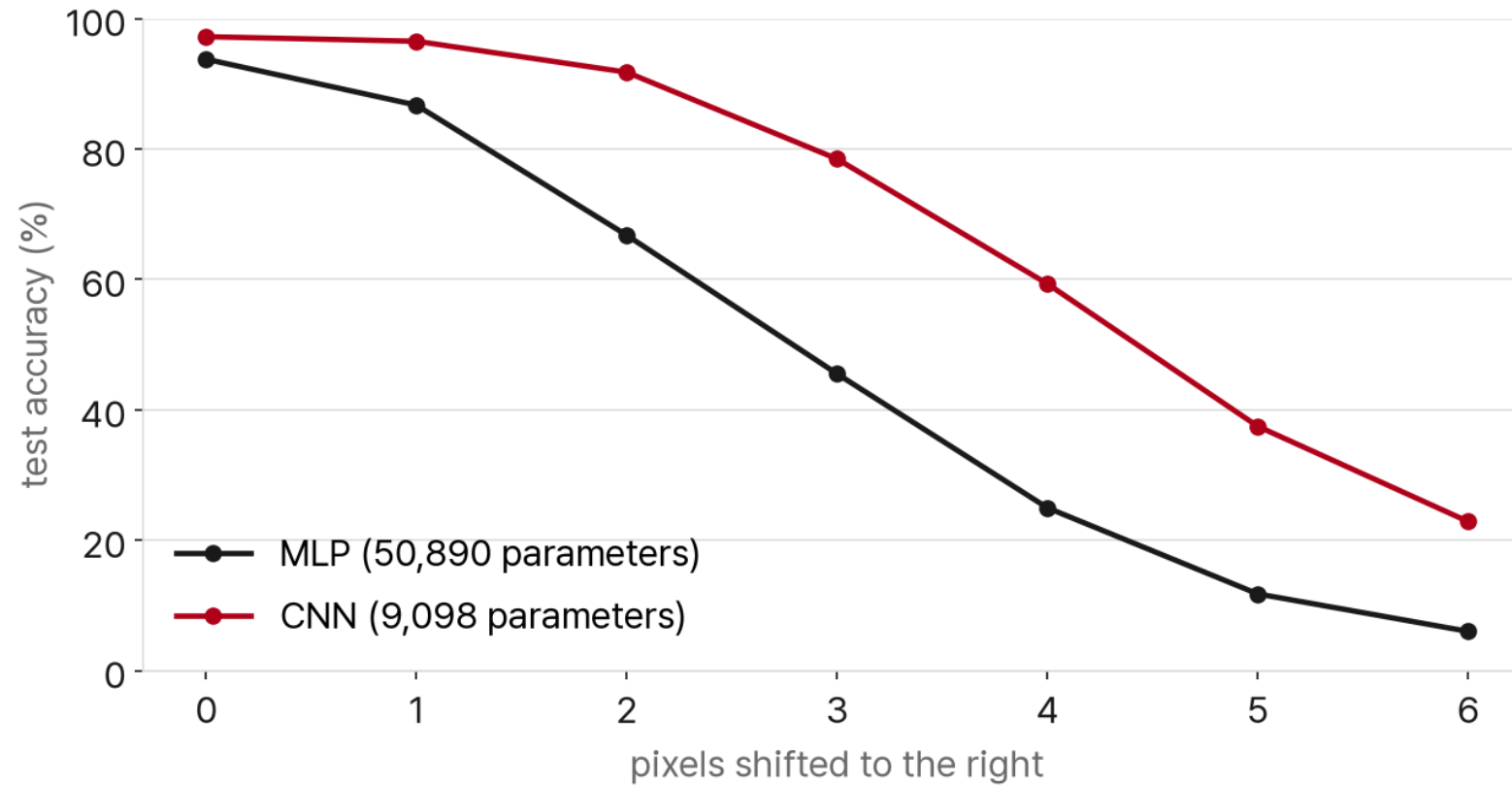
Layer	Output shape
Input	$1 \times 28 \times 28$
8 filters of $1 \times 3 \times 3$, ReLU	$8 \times 28 \times 28$
Max pooling 2×2	$8 \times 14 \times 14$
16 filters of $8 \times 3 \times 3$, ReLU	$16 \times 14 \times 14$
Max pooling 2×2	$16 \times 7 \times 7$
Flatten, then 784×10	10

Counting the Parameters

Layer	Parameters
8 filters of $1 \times 3 \times 3$, plus 8 biases	80
16 filters of $8 \times 3 \times 3$, plus 16 biases	1,168
784×10 , plus 10 biases	7,850
CNN total	9,098
MLP from lecture 7	50,890

Test Accuracy After Shifting

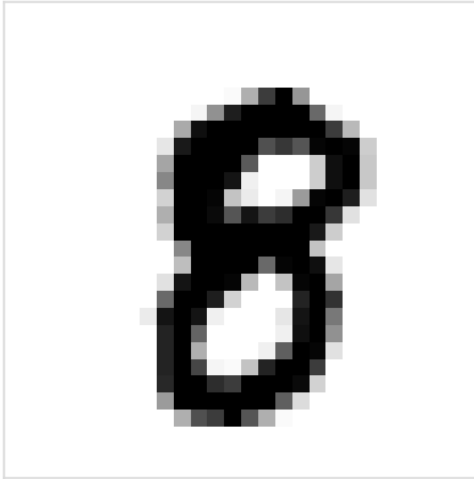
All 2,000 test images, moved the same number of pixels



Shifting the Digit

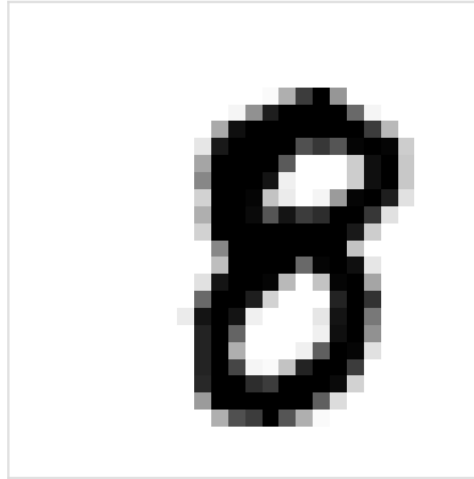
The trained CNN, the same test image

shifted 0 px right



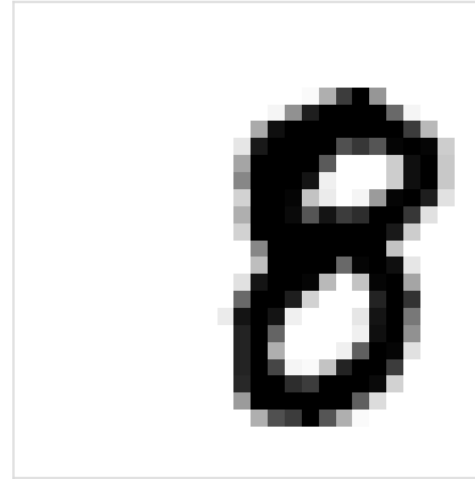
answer 8 $p(8) = 1.00$

shifted 2 px right



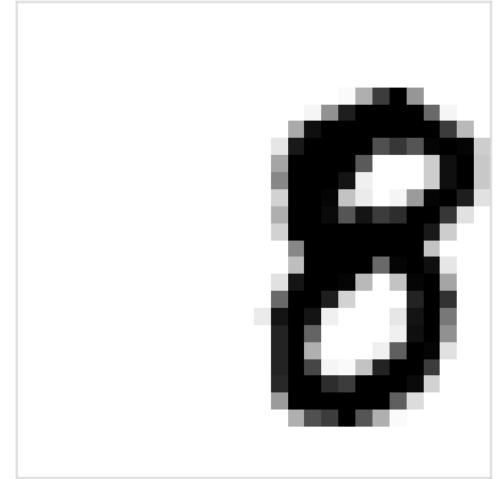
answer 8 $p(8) = 1.00$

shifted 4 px right



answer 8 $p(8) = 0.95$

shifted 6 px right



answer 4 $p(8) = 0.01$

ResNet18

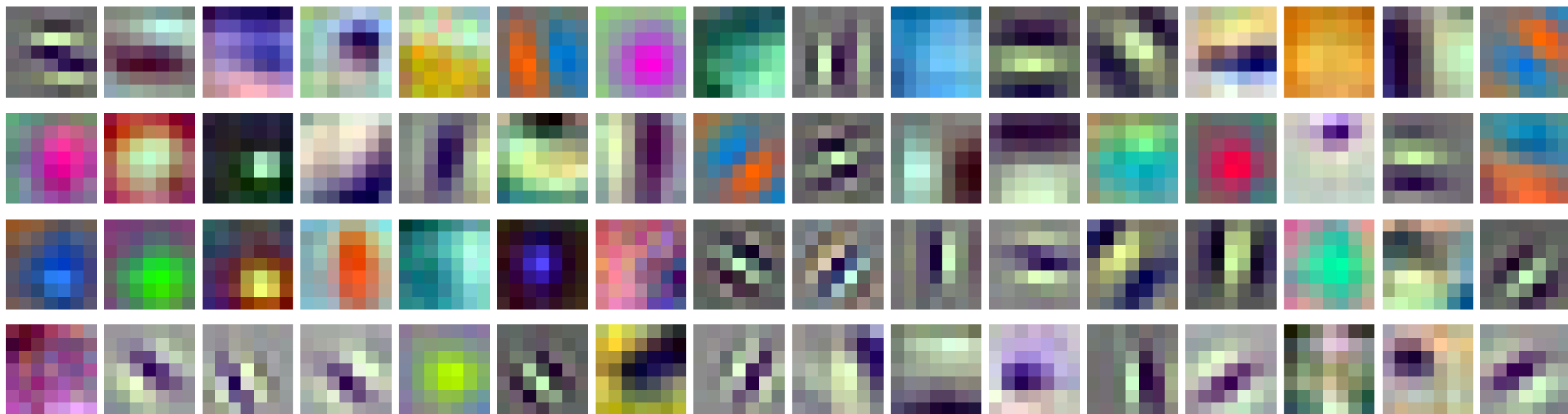
ResNet18

He et al. (2016), trained on ImageNet: 1.28 million photos, 1,000 categories

- Input $3 \times 224 \times 224$, output 1,000 scores
- 20 convolution layers, 11,689,512 parameters
- Assignment 1 replaces the last layer: $512 \times 1,000$ becomes $512 \times \text{your categories}$

First-Layer Filters of ResNet18

64 filters of $3 \times 7 \times 7$, after training on ImageNet



Coming Up

- Assignment 1 due Friday, October 16, 22:00
- Next: starting from a network that was already trained